
PROJECT TITLE

Guarding Transactions with AI-Powered Credit Card Fraud Detection and Prevention Using AWS Cloud

CHAPTER 1: INTRODUCTION

The increasing use of online transactions and digital payment systems has significantly improved financial accessibility. However, it has also increased credit card fraud incidents worldwide. Traditional fraud detection systems use static rules that struggle to identify sophisticated fraud patterns.

This project develops an AI-powered cloud-based fraud detection system that uses Machine Learning and AWS services to identify suspicious transactions in real time. The system stores data securely, processes transactions automatically, and generates alerts when fraudulent activities are detected.

Objectives:

- Detect fraudulent transactions in real time.
- Improve transaction security.
- Reduce financial losses.
- Automate fraud alerts.
- Build a scalable cloud architecture.

CHAPTER 2: LITERATURE SURVEY

Existing systems:

- Rule-based fraud detection
- Manual transaction monitoring
- Statistical analysis methods

Limitations:

- Unable to adapt to new fraud patterns
- High false positive rates
- Limited scalability

Proposed improvements:

- Machine Learning integration
- Cloud automation
- Real-time processing
- Automated alert systems

CHAPTER 3: SYSTEM REQUIREMENTS

Hardware Requirements:

- Processor: Intel i5 or above
- RAM: 8 GB minimum

- Storage: 20 GB free space
- Internet connection

Software Requirements:

- Python 3.11+
- VS Code
- AWS Account
- Pandas
- NumPy
- Scikit-Learn
- Joblib

AWS Services:

- S3
- Lambda
- DynamoDB
- SNS
- API Gateway
- CloudWatch

CHAPTER 4: DATASET COLLECTION AND PREPROCESSING

Dataset Source:

Kaggle Credit Card Fraud Detection Dataset

Preprocessing Steps:

- Dataset loading
- Missing value verification
- Duplicate removal
- Fraud distribution analysis
- Feature preparation
- Clean dataset generation

Files:

- creditcard.csv
- clean_creditcard.csv

CHAPTER 5: MACHINE LEARNING MODEL DEVELOPMENT

Algorithm:

Random Forest Classifier

Steps:

- Load dataset
- Separate features and labels
- Split training and testing data

- Train model
- Evaluate performance
- Save fraud_model.pkl

Performance Metrics:

- Accuracy
- Precision
- Recall
- F1 Score

CHAPTER 6: AWS CLOUD IMPLEMENTATION

Services Used:

Amazon S3:

Stores datasets and model files.

AWS Lambda:

Processes transactions.

Amazon DynamoDB:

Stores transaction history.

Amazon SNS:

Sends email alerts.

Amazon API Gateway:

Provides API endpoints.

CloudWatch:

Monitors application logs.

CHAPTER 7: SYSTEM ARCHITECTURE

Transaction Flow:

User



API Gateway



Lambda



Fraud Detection



DynamoDB



SNS Alerts



Dashboard

CHAPTER 8: IMPLEMENTATION PROCESS

Phase 1:

Project Planning

Phase 2:

AWS Setup

Phase 3:

Dataset Preparation

Phase 4:

Model Training

Phase 5:

AWS Integration

Phase 6:

Testing and Deployment

CHAPTER 9: RESULTS AND OUTPUT

Outputs:

- Clean dataset generated.
- Fraud model created.
- Transaction records stored.
- Email notifications generated.
- Fraud detection executed successfully.

Advantages:

- Fast processing
- High scalability
- Automated detection
- Secure cloud storage
- Industry-level implementation

CHAPTER 10: CONCLUSION AND FUTURE ENHANCEMENTS

Conclusion:

The project successfully demonstrates the integration of Machine Learning and AWS cloud services for real-time fraud detection. The system improves financial security by automating fraud identification and notification processes.

Future Enhancements:

- Mobile application integration
- Advanced deep learning models
- Multi-bank support
- Real-time dashboards
- Multi-cloud deployment